A 2D-Extendable 48V-1V Ladder-Series-Capacitor Buck Converter for Computing Power Delivery

Zhiguo Tong[1,2], Wenjie Yang[1], Zhewen Yu[1], Shousheng Han[2], Rui P. Martins[2], Yan Lu[1]
[1]Tsinghua University, [2]University of Macau

2026 IEEE/JSAP Symposium on VLSI Technology and Circuits (VLSI Technology and Circuits) | 979-8-3315-8096-4/26/$31.00 ©2026 IEEE | DOI: 10.1109/VLSITECHNOLOGYANDCIR65830.2026.11577585

Abstract

This paper presents a 48V-to-1V two-dimensional extendable ladder-series-capacitor buck converter (LSCB) which supports both input-voltage stackable and output-current parallel architecture of power cells, to reduce switch voltage stress and extend output current simultaneously. We also implement an input-series output-parallel (ISOP) two-stage configuration to realize vertical power delivery. Fabricated in 180-nm BCD for input-series chips and 65-nm process for output-parallel chips, the cascaded ISOP system achieves a peak power density of 2132 W/in³ while delivering a maximum output current of 25 A.

Introduction

The rapid growth of AI workloads has imposed stringent requirements on voltage regulators (VRs), including large output current, elevated input voltage, wide load range, and fast transient response [1]. As the required power level continues to increase far beyond the capability of a single converter, the ability to handle high input voltage and large output current has become a fundamental requirement for modern VRs (Fig. 1).

Conventional approaches address this requirement through either parallel or stackable structures. Parallel phase extension, as in multiphase buck converters, effectively increases output current and supports phase shedding [2]. However, under high input voltage, the switch voltage stress increases directly with the bus voltage, limiting efficiency and power density. Hybrid architectures such as the hybrid dual-ring (HOOP) structure [3] improve current sharing but do not further reduce switch voltage stress with increasing phase count. Alternatively, stackable architectures such as switched-capacitor buck converters reduce device voltage stress through voltage stacking. Nevertheless, their narrowed conversion-ratio range and non-self-maintained flying-capacitor voltages complicate phase interleaving and phase shedding, making it difficult to support high output current and wide load-range operation.

To overcome the inherent trade-off between current scaling and voltage-stress reduction, this work proposes a two-dimensional (2D) extendable architecture that combines phase parallel and stage stacking within a unified framework (Fig. 2). Compared with conventional parallel structures, the switch voltage stress decreases with the number of stacked stages. Compared with traditional stackable converters, the flying-capacitor voltages are maintained during idle operation, and all inductors are energized concurrently, enabling phase shedding and fast transient response. As a result, the proposed 2D-extendable architecture is well suited for high-current, high-voltage VRs in advanced CPD applications.

Proposed 2D Extendable LSCB Converter

To demonstrate the proposed concept, the ladder-series-capacitor buck (LSCB) converter is presented in Fig. 3 (left), where a four-stage implementation is shown as an example. The LSCB employs four DC capacitors, C_{DCX}, to divide the high input voltage into four low-voltage domains, each supplying a dedicated power-conversion cell for energy transfer to the output. In steady-state operation, the LSCB behaves similarly to a conventional switched-capacitor buck (SCB) converter, as illustrated in Fig. 4 (left). The flying-capacitor voltages V_{CF2}, V_{CF3}, and V_{CF4} naturally settle at approximately $V_{IN}/4$, $V_{IN}/2$, and $3V_{IN}/4$, respectively. Owing to the intrinsic body diodes of S_{DC}, the DC-capacitor voltages are automatically balanced to $V_{IN}/4$. As a result, the LSCB inherits the key advantages of SCB converters, including soft charging of capacitors and reduced voltage stress. Beyond steady-state operation, the LSCB offers enhanced transient and idle-state performance. During load transients, all inductors are energized simultaneously, resulting in more than a 2× increase in the aggregate output current slew rate. In the idle state, shown in Fig. 4 (right), the C_{DCX} capacitors clamp each flying-capacitor voltage C_{FX} to its nominal level, enabling phase shedding and supporting a wide load range. To further enable vertical power delivery for computing systems, a three-phase interleaved 3:1 Dickson switched-capacitor (SC) converter is cascaded with the LSCB module to form an input-series output-parallel (ISOP) architecture (Fig. 3). The low-profile SC stage is embedded beneath the substrate, enabling vertical power delivery while providing an elevated intermediate voltage that improves power transmission efficiency and reduces distribution losses [6].

Fig. 5 shows the block diagram of the proposed two-stage ISOP system and highlights two key circuit techniques: a soft-start pre-charging scheme and a floating bootstrap gate driver. To ensure safe startup under a 48-V input, a resistor-based pre-charging circuit is employed to initialize both DC and flying capacitors without incurring steady-state loss (Fig. 6). In addition, to drive stacked power cells without excessive gate-drive voltage drop, a floating bootstrap gate-drive scheme is adopted. In this scheme, the gate-drive voltage is generated from the floating $V_{IN}/4$ rail using an LDO-based regulator followed by a 2:1 switched-capacitor converter to improve conversion efficiency (Fig. 6). Measurement results confirm the generation of a 29-V floating gate-drive voltage from a 36-V source. Furthermore, a transient-detection circuit is incorporated to enable fast transient response.

Measurement Results

Fig. 7 presents the implemented four-stage LSCB chip fabricated in a 180-nm BCD process, and the Dickson SC implemented in a 65-nm CMOS process. Fig. 8 shows the measurement board, where one LSCB converter is cascaded with four Dickson SC converters. Zero-ohm resistors allow individual SC converters to be bypassed for evaluation. The assembled vertical power-delivery demonstration module occupies a compact volume of 5.5 × 6.5 × 4.3 mm³. Fig. 9 presents the measured waveforms during start-up, steady-state operation, and mode transition. As V_{IN} ramps from 0 to 48 V, all DC and flying capacitors are pre-charged to their nominal voltages. In steady state, the V_{LX1}–V_{LX4} nodes toggle between 0 and 12 V, confirming voltage stress reduction. The transition between steady and idle state demonstrates instant phase shedding enabled by the flying-capacitor voltage clamping.

Fig. 10 presents the measured waveforms, efficiency, and performance comparison of the proposed system. Owing to the interleaved operation of the SC stage, only 31 mV ripple is observed at the intermediate node without a mid-capacitor. A peak efficiency of 91.2% is achieved with a single-phase LSCB and 90.7% with a two-phase LSCB at output voltage of 1 V. The two-phase LSCB supports up to 20 A output current with an efficiency of 87.6%. Comparisons of peak efficiency and peak current density demonstrate a state-of-the-art trade-off compared to other 48V-to-1V converters. Table I summarizes this work and compares it with recent designs [7], [8]. The proposed architecture introduces a 2D extendable structure that enables simultaneous handling of high input voltage and large output current. The two-phase, four-stage LSCB delivers up to 20 A with a peak efficiency of 91.7%. The ISOP system achieves a peak output current of 25 A with a peak power density of 2132 W/in³.

Acknowledgement: This work is supported by National Natural Science Foundation of China under Grant 92573201.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on October 30,2026 at 09:57:55 UTC from IEEE Xplore. Restrictions apply.

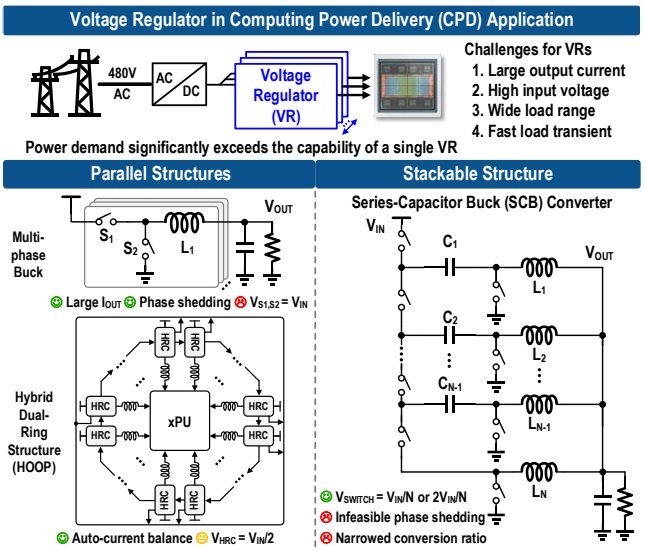


Fig. 1. The challenges for VRs in CPD application, and conventional parallel and stackable structures and their pros and cons.

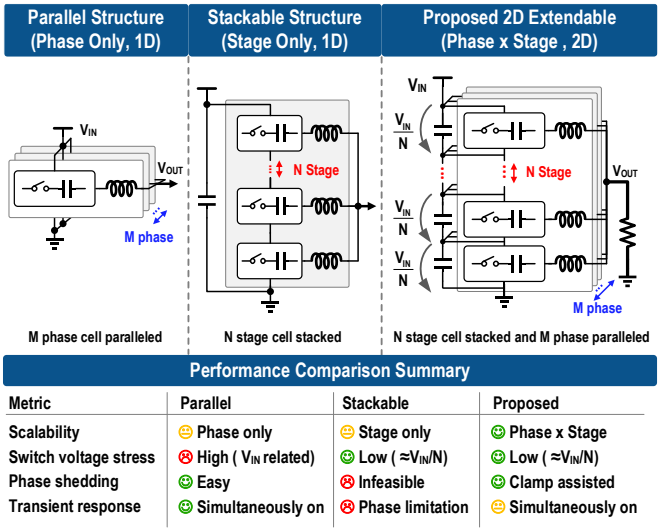


Fig. 2. The summary and comparison of the parallel, stackable and the proposed 2D extendable structure.

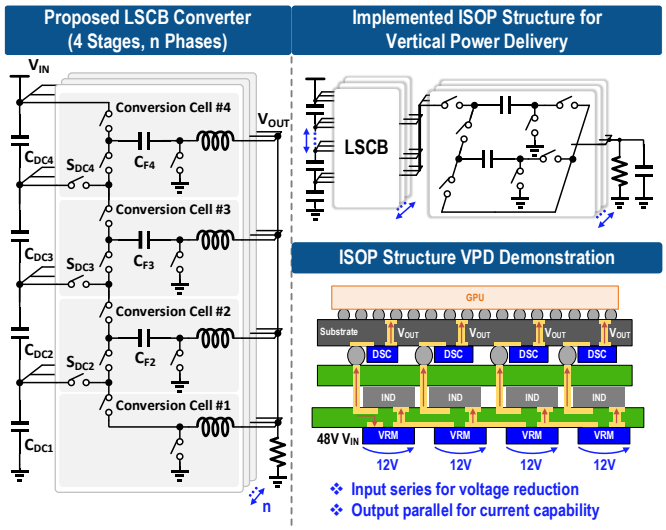


Fig. 3. The proposed LSCB converter with four stages and input-series output-parallel (ISOP) two-stage architecture for vertical power delivery and its 3D demonstration.

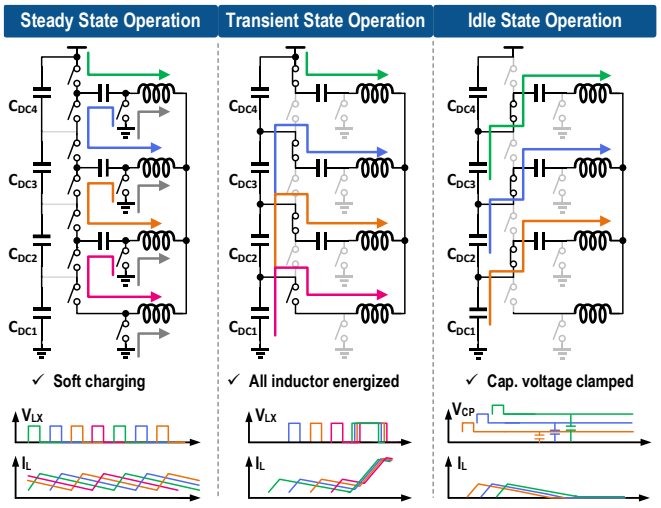


Fig. 4. The operation principle of the steady state, transient state and idle state of LSCB converter.

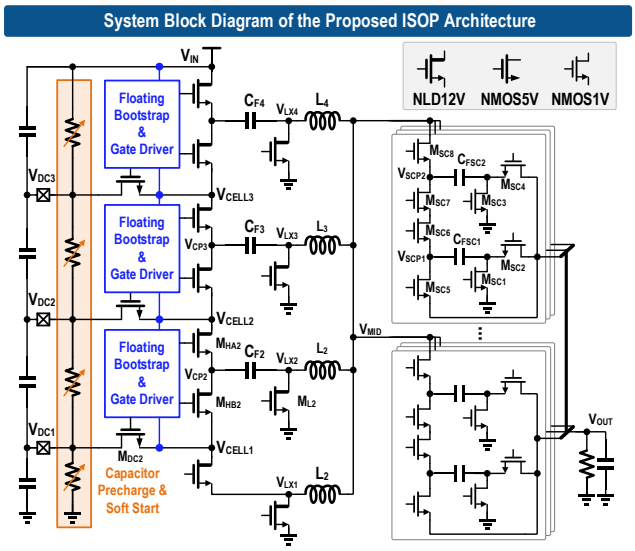


Fig. 5. The power stage block diagram of the ISOP two-stage architecture.

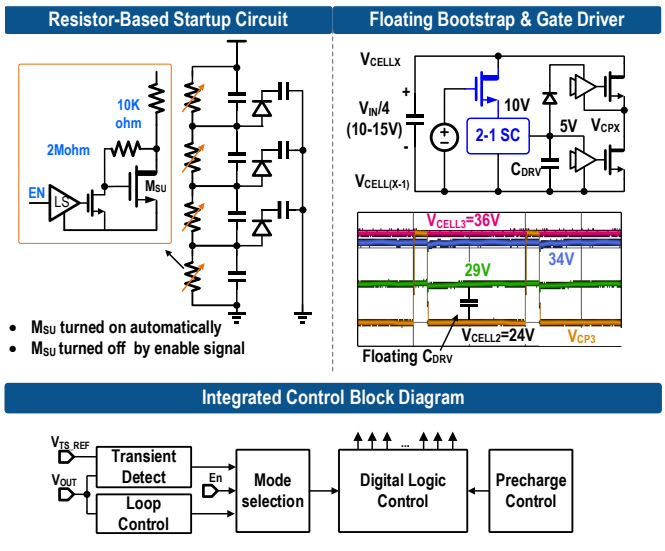


Fig. 6. The resistor-based startup circuit and floating bootstrap and gate driver technique.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on October 30,2026 at 09:57:55 UTC from IEEE Xplore. Restrictions apply.

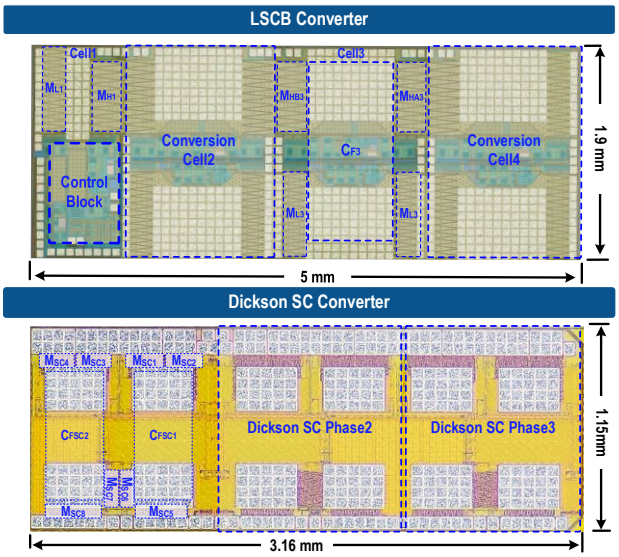


Fig. 7. Chip micrograph of the LSCB converter (top), and Dickson SC converter (bottom).

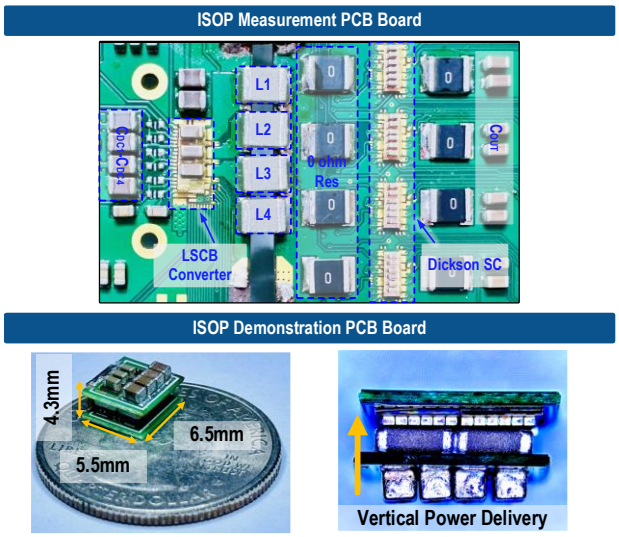


Fig. 8. The ISOP measurement PCB board (top) and ISOP demonstration PCB board (bottom).

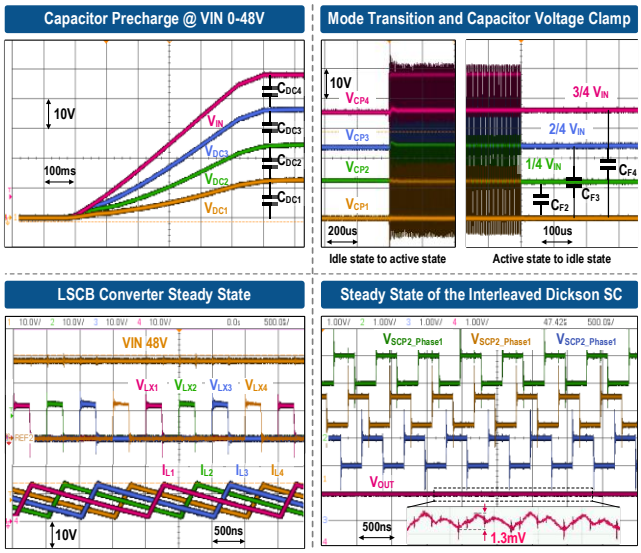


Fig. 9. Measured start-up waveforms (top left), mode transition and voltage clamp (top right), steady state of LSCB (bottom left) and Dickson SC (bottom right).

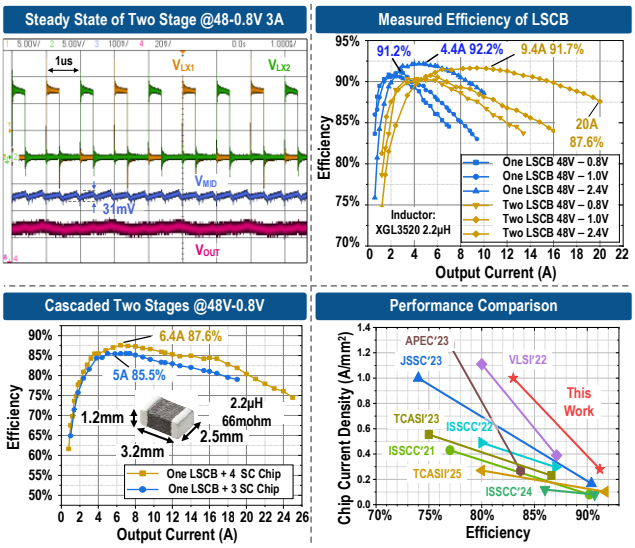


Fig. 10. Measured cascaded two-stage waveforms (top left), efficiency of LSCB (top right), efficiency of two stage (bottom left) and performance comparison (bottom right).

TABLE I. The comparison table with the state-of-the-art works.

	ISSCC' 21	VLSI'22	JSSC'23	ISSCC'24	This Work	
Technology	0.18μm BCD	0.18μm BCD	0.18μm BCD	N.A.	0.18μm BCD + 65nm	
Structure	12-level series-capacitor buck	QSD	3-level hybrid dickson	Buck + SPCSC	LSCB	LSCB + SC
Scalability	Stacked	Stacked	Parallel	Parallel	2D Extendable	
V_{IN}	36V-60V	48V	48V	12V-60V	40V-56V	
V_{OUT}	0.5V-1V	1V / 1.2V	0.7V-1V	1.2	0.8V-2.4V	0.6V-0.8V
Maximum Output Current	8A	10A	12A	2A	10A	25A
Switching Frequency	5MHz	500KHz/1MHz	1MHz	500KHz	500KHz	500KHz + 1.6MHz
Inductor	2x 0.1μH	4×1μH	3 x 0.62μH	4 x 10μH	4 x 2.2μH	
Inductor Location	Output	Output	Output	Middle	Output	Middle
Number of External Switches	1	0	2	0	0	0
Peak Efficiency	90.2% @48/1	87.1@48/1	90.4% @48/1	90.7% @60/1.2	91.2%@48/1	87.6%@48/0.8
Chip Current Density	0.44 A/mm^2	1.11 A/mm^2	1 A/mm^2	0.12 A/mm^2	1 A/mm^2	
Current Density	998W/inch^3	N.A.	992W/inch^3	N.A.	996 W/inch^3	2132 W/inch^3

References

[1] K. Radhakrishnan *et al.*, *IEEE TCPMT*, vol. 11, no. 4, pp. 655–671, 2021, doi: 10.1109/TCPMT.2021.3065690.

[2] C. Huang *et al.*, *IEEE JSSC*, vol. 48, no. 12, pp. 2977–2988, 2013, doi: 10.1109/JSSC.2013.2286545.

[3] Z. Tong *et al.*, ISSCC, 2025, pp. 1–3, doi: 10.1109/ISSCC49661.2025.10904695.

[4] X. Yang *et al.*, ISSCC, 2021, pp. 466–468, doi: 10.1109/ISSCC42613.2021.9366005.

[5] H. Han *et al.*, Symp. on VLSI, 2022, pp. 182–183, doi: 10.1109/VLSITechnologyandCir46769.2022.9830233.

[6] C. Hardy *et al.*, ISSCC, 2023, pp. 182–184, doi: 10.1109/ISSCC42615.2023.10067315.

[7] M. Gong *et al.*, IEEE JSSC, vol. 58, no. 4, pp. 1002–1014, 2023, doi: 10.1109/JSSC.2022.3228233.

[8] H.-J. Choi *et al.*, ISSCC, 2024, pp. 156–158, doi: 10.1109/ISSCC49657.2024.10454344.

Authorized licensed use limited to: BEIHANG UNIVERSITY. Downloaded on October 30,2026 at 09:57:55 UTC from IEEE Xplore. Restrictions apply.